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A Small MSCI World GARCH Tournament

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1 Introduction

The aim of this project is the direct application of the knowledge acquired in the study of the Financial Econometrics course material with special regard to volatility modeling and forecasting and risk assessment. The choice of the MSCI World Index (via `URTH` ETF) as a case study is rather arbitrary and mostly driven by a simple matter of personal curiosity. Volatility, Value-at-Risk (VaR) and Expected Shortfall (ES) forecasts are computed for 1d, 5d and 10d horizons and validated against an out of sample time window of data.

In the **In-Sample Analysis** section preliminary diagnostics of the time series are carried out to assess the distributional structure of returns over its entire history. The presence of ARCH effect is then tested, so as to justify the later use of GARCH-family models. Said models are estimated via maximum likelihood, with Student-t innovations and various diagnostics are carried out on the resulting residuals.

In **Out-of-Sample Analysis** section EWMA, GARCH, GJR-GARCH and EGARCH out-of-sample forecasts are compared via Diebold-Mariano and Giacomini-White tests. 95% and 99% confidence VaR and ES are then computed for GARCH, EGARCH and GJR-GARCH, under both normal and student-t innovations, and a naive Historical Simulation at all 3 time horizons. VaR conditional coverage is also tested.

The daily closing price data from 02/01/2013 to 03/09/2026 of the iShares MSCI World UCITS ETF (`URTH`) are fetched via the yfinance (Yahoo Finance) library totaling 3439 closings and 3438 daily returns after differencing. Returns are specified as percentage log,

$$r_t = 100 \times \ln\left(\frac{P_t}{P_{t-1}}\right), \tag{1}$$

with P_t closing price on day t, as is the convention.

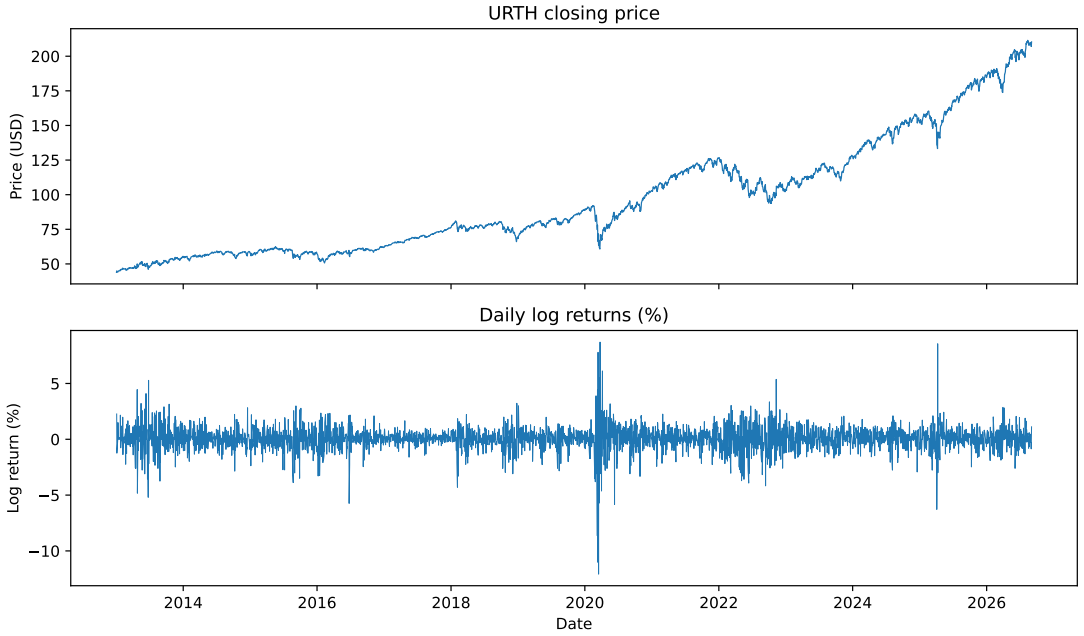


Figure 1: Full sample plot. **Top:** URTH closing price. **Bottom:** percentage daily log returns.

2 In-Sample Analysis

2.1 General Statistics

A preliminary analysis of the return distribution is carried out over the entire time series yielding a mean of 0.0455, a standard deviation of 1.0844, a skewness of -0.764 and an excess kurtosis of 14.569. One can already appreciate that the distribution is slightly left-skewed and the evidently heavy-tailed but for parsimony a Jarque-Bera test **[9, 10]** is carried out against the null hypothesis of normality $JB \sim \chi^2_2$. The test statistic is

$$JB = \frac{n}{6}\left(S^2 + \frac{K^2}{4}\right) \tag{2}$$

where n is the number of data, S is the skewness and K is the excess kurtosis. The statistic turns out to be 30738.3 ($p << 0.0001$), mainly due to the contribution of the excess kurtosis, which leads to the overwhelmingly clear rejection of the hypothesis of normality. Given this results the in-sample analysis is carried out using Student-t innovations instead of normal. As per the out-of-sample analysis both are studied so as to assess the hypothetical underestimation of tail risk under normality.

2.2 Testing for ARCH Effects

A first qualitative assessment of the presence of ARCH effects can be drawn from the evidently slow decay of the autocorrelation function (ACF) in **Figure 2**. The absence of a hard cut-off after a given lag is already heuristic evidence of the presence of volatility clustering in the return series. It is however more appropriate to statistically test the presence of ARCH effects **[6]** which is done via Ljung-Box test **[3, 12]** and ARCH-LM test **[6]** before fitting any GARCH-type model on our data.

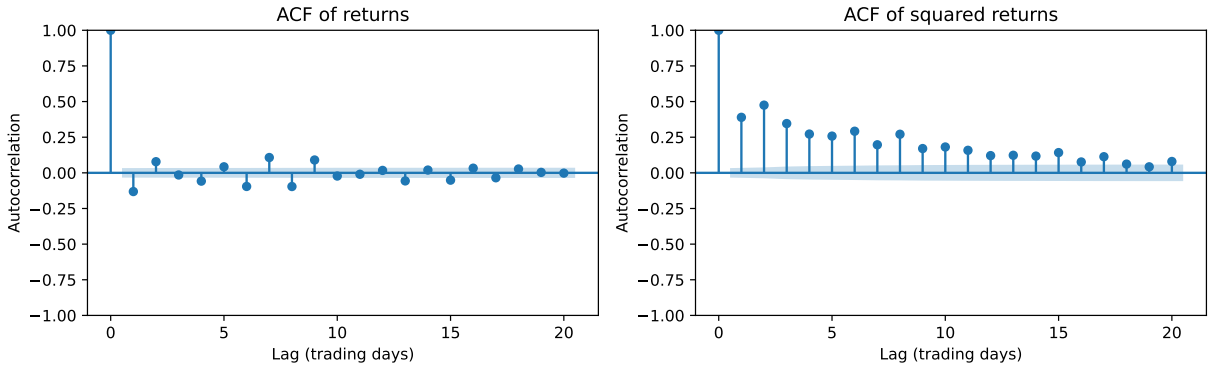


Figure 2: Left: Autocorrelation function of level returns. Right: Autocorrelation function of squared returns. 20 lags.

The LB statistic on squared return is

$$Q(k) = T(T+2)\sum_{j=1}^{k}\frac{\hat{\rho}_j^2}{T-j} \tag{3}$$

with $\hat{\rho}_j = \frac{\sum_{t=j+1}^{T}(r_t^2-\bar{r^2})(r_{t-j}^2-\bar{r^2})}{\sum_{t=1}^{T}(r_t^2-\bar{r^2})^2}$ the autocorrelation of squared returns, k the number of lags considered and T the length of the series. The statistic is tested against the null of no autocorrelation up to lag k, formally $H_0: \; Q(k) \sim \chi^2_k$.

The objective of the ARCH-LM test proposed by Engle is to test the autoregressive nature of squared residuals on the conditional mean (here taken as constant)

$$\hat{\varepsilon}_t^2 = \gamma_0 + \sum_{j=1}^{k}\gamma_j\,\hat{\varepsilon}_{t-j}^2 + u_t \tag{4}$$

with null of no ARCH effect $H_0: \; \gamma_1 = \cdots = \gamma_k = 0$ which can be rejected if a single one of the k condition is violated. The LM statistic is computed as

$$LM = T \cdot R^2, \tag{5}$$

where R^2 is the coefficient of determination of the autoregression on the squared residuals, and tested against null χ^2_k.

By testing for both 10 and 20 lags we obtain $LB_{sq,10} = 3086.24$ ($p < 0.0001$), $LB_{sq,20} = 3498.20$ ($p < 0.0001$), $LM_{10} = 1047.66$ ($p < 0.0001$) and $LM_{20} = 1080.92$ ($p < 0.0001$) which leads to a rejection of both nulls on both horizons, meaning that the returns actually exhibit ARCH effects and the adoption of GARCH-type models is legitimate.

2.3 In-Sample Fit

Three well known GARCH-type models with Student-t innovations are estimated via maximum likelihood on the whole series. The first subject of the study is the standard GARCH(1,1) **[2, 6]**

$$\sigma_t^2 = \omega + \alpha\,\varepsilon_{t-1}^2 + \beta\,\sigma_{t-1}^2, \tag{6}$$

and it is confronted against the GJR-GARCH(1,1) specification **[8]**

$$\sigma_t^2 = \omega + \alpha\,\varepsilon_{t-1}^2 + \gamma\,\mathbb{1}_{[\varepsilon_{t-1}<0]}\,\varepsilon_{t-1}^2 + \beta\,\sigma_{t-1}^2 \tag{7}$$

which adds an additional asymmetric term so as to raise volatility more under negative rather than positive shocks. The third observed model is the EGARCH(1,1) **[14]**, which captures the shocks asymmetry via the log specification itself,

$$\ln\sigma_t^2 = \omega + \alpha\big(|z_{t-1}| - \mathbb{E}|z_{t-1}|\big) + \gamma\,z_{t-1} + \beta\,\ln\sigma_{t-1}^2 \tag{8}$$

where $z_{t-1} = \varepsilon_{t-1}/\sigma_{t-1}$ is the standardized shock or innovation.

The estimated parameters are reported in **Table 1** where it can also be seen that it is not possible to reject in good confidence the compatibility of α for GJR-GARCH-t and ω for EGARCH-t with zero, suggesting that said estimates are not really significant. This means that for GJR-GARCH-t the response to shocks is fully captured by the asymmetric term and very little by the symmetric one. The estimated values of ν further highlight the presence of fat-tails in the data. The fitted models are illustrated in **Figure 3**.

Table 1: Parameters estimated via MLE under Student-t innovations.

Model	Parameter	Estimate	Std. Error	t-stat	p-value
GARCH-t	μ	0.088	0.012	7.637	< 0.0001
	ω	0.0242	0.0062	3.928	< 0.0001
	α	0.155	0.020	7.830	< 0.0001
	β	0.833	0.019	42.898	< 0.0001
	ν	5.65	0.53	10.735	< 0.0001
GJR-GARCH-t	μ	0.062	0.012	5.297	< 0.0001
	ω	0.0252	0.0053	4.767	< 0.0001
	α	0.011	0.013	0.843	0.3990
	γ	0.235	0.033	7.204	< 0.0001
	β	0.851	0.018	47.445	< 0.0001
	ν	5.94	0.60	9.860	< 0.0001
EGARCH-t	μ	0.057	0.012	4.868	< 0.0001
	ω	-0.0051	0.0045	-1.147	0.2510
	α	0.166	0.025	6.766	< 0.0001
	γ	-0.165	0.015	-11.049	< 0.0001
	β	0.9714	0.0060	161.913	< 0.0001
	ν	5.96	0.61	9.849	< 0.0001

The models' fits are then ranked by Akaike **[1]** and Bayesian **[15]** Information Criteria (AIC and BIC), in order to assess the fit-complexity tradeoff. The two criteria are

$$\mathrm{AIC} = -2\ln\mathcal{L} + 2k, \tag{9}$$

$$\mathrm{BIC} = -2\ln\mathcal{L} + k\ln n, \tag{10}$$

where $\mathcal{L}$ is the maximized likelihood of the model, capturing the goodness of the fit, k the number of parameters, capturing the complexity of the model, and n is the sample size, which makes the BIC penalize the complexity more than the AIC. The criteria results are reported in **Table 2** in decreasing AIC/BIC rank order. The main

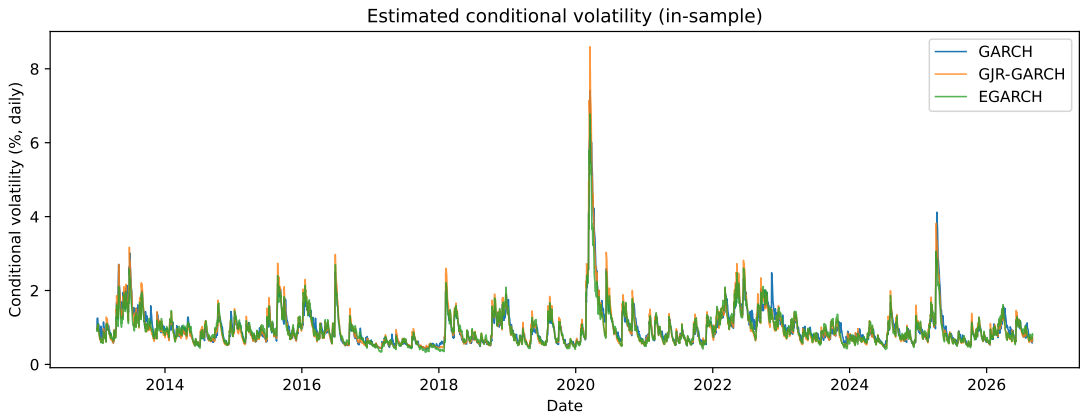


Figure 3: Estimated conditional volatility for GARCH-t, GJR-GARCH-t and EGARCH-t on the full series.

result that can be appreciate from this comparison is that the increase in complexity of the models does pay off and regardless of the complexity penalization EGARCH-t stands out of the three as the model that offers the best fit-complexity tradeoff although the relative differences in performance are small.

Table 2: In-sample model comparison.

Model	Log-Likelihood	AIC	BIC
GARCH-t	-4359.88	8729.77	8760.48
GJR-GARCH-t	-4313.28	8638.55	8675.41
EGARCH-t	-4301.55	8615.09	8651.95

2.4 Residual Diagnostics

A good AIC/BIC ranking is not per se evidence of the model adequateness. The point of fitting a GARCH-type model is to capture the ARCH effects on the original return series documented in **Section 2.2**, as such the Ljung-Box and ARCH-LM tests are re-run on each model's standardized residuals $\hat{z}_t = \hat{\varepsilon}_t / \hat{\sigma}_t$. However, unlike what presented in **Section 2.2**, a high p-value here is the expected good outcome for it indicates the absence of significant remaining autocorrelation. The results of the LB and ARCH-LM tests are reported in **Table 3**. To complement the tests a QQ-plot of the standardized residuals is presented in **Figure 4**. The plot shows whether the assumed innovation distribution (Student-t) is a reasonable description of the data.

Table 3: All p-values from the residual diagnostics on standardized residuals.

Model	$LB_{lvl,10}$	$LB_{lvl,20}$	$LB_{sq,10}$	$LB_{sq,20}$	LM_{10}	LM_{20}
GARCH-t	0.0822	0.1001	0.2048	0.4357	0.2028	0.3766
GJR-GARCH-t	0.0154	0.0149	0.5004	0.8150	0.5010	0.7832
EGARCH-t	0.0026	0.0050	0.1901	0.6088	0.1828	0.5393

From the p-values on the LB test on squared residuals and the ARCH-LM test it is possible to state in good confidence that ARCH effects, previously observed on the raw data, have effectively been removed by the models. It can also be seen that the LB test fails on level residuals for GJR-GARCH and EGARCH suggesting that the constant mean specification is too rigid for them.

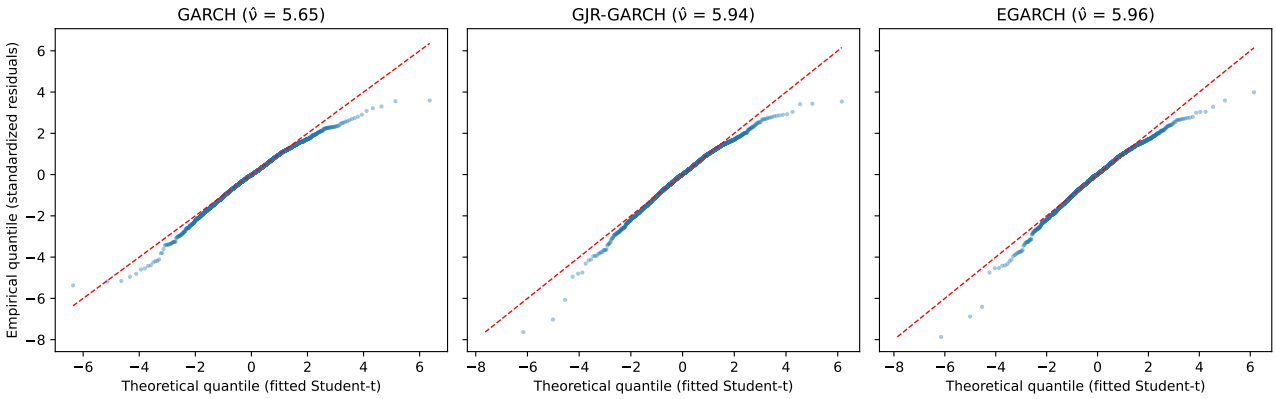


Figure 4: QQ-plot of standardized residuals against each model's fitted Student-t distribution.

The QQ-plot in **Figure 4** shows an asymmetric departures of the tails from the Student-t distribution. The fit is good in the central quantiles, roughly from -2 to 2, for all models. The visible asymmetry is related to the skewness observed in **Subsection 2.1** and is shared by all models, further suggesting that a future extension to the model comparison should include asymmetric innovation distributions. The plot also suggests that the models should not prove to be too pathological in the context of Value-at-Risk estimation and forecasting but could prove to be more of an issue for what concerns the Expected Shortfall as it averages the entirety of the tails, including poorly fit extremes.

3 Out-of-Sample Analysis

3.1 Forecasting

For the out-of-sample comparison each model is estimated on a 1000-day rolling window which is re-estimated every 20 trading days. Between re-estimations the conditional-variance recursion is updated daily. Forecasts are then evaluated at 1, 5, and 10 trading days horizons.

For further reference the three models are also compared against the Exponentially Weighted Moving Average (EWMA) with fixed hyperparameter λ:

$$\hat{\sigma}_t^2 = \lambda\,\hat{\sigma}_{t-1}^2 + (1-\lambda)\,r_{t-1}^2, \qquad \lambda = 0.94, \tag{11}$$

which is taken as a naive benchmark as it does not require any parameter estimation.

The cumulative h-day ($h = 1, 5, 10$) ahead variance forecast is computed as

$$\widehat{\Sigma}_{t,h} = \sum_{k=1}^{h} \mathbb{E}_t\left[\sigma_{t+k}^2\right] \tag{12}$$

and evaluated against a realized variance proxy

$$\widehat{\Sigma}_{t,h}^{\text{real}} = \sum_{k=1}^{h} (r_{t+k} - \hat{\mu}_t)^2 \tag{13}$$

via MSE and QML loss functions

$$\text{MSE} = \frac{1}{T}\sum_t \left(\widehat{\Sigma}_{t,h}^{\text{real}} - \widehat{\Sigma}_{t,h}\right)^2, \tag{14}$$

$$\text{QML} = \frac{1}{T}\sum_t \left[\ln \widehat{\Sigma}_{t,h} + \frac{\widehat{\Sigma}_{t,h}^{\text{real}}}{\widehat{\Sigma}_{t,h}}\right]. \tag{15}$$

The forecast results pictured in **Figure 5** and the computed losses are reported in **Table 4**.

Beside observing that EWMA under performs all GARCH-type models on both metrics on all horizons, except for 10d EGARCH QML, which suggests that the added complexity of said models is fruitful, it is important to observe that the MSE and QML rankings disagree with one another. This is expected as the MSE loss is symmetric and more sensible to extreme events whilst the QML loss is asymmetric and more robust to the extremes which makes of it the usually preferred loss for this kind of application.

3.2 Forecast comparison

The statistical significances of various models' forecasts' differences are compared via Diebold-Mariano (DM) test [5], valid only for non-nested models, and Giacomini-White (GW) test [7], valid instead for nested models. Specifically GARCH and GJR-GARCH are nested as GJR-GARCH collapses to GARCH for $\gamma = 0$, EWMA is nested with GARCH as it is a GARCH with $\omega = 0$ and $\alpha + \beta = 1$, EWMA and GJR-GARCH are tansitively nested and EGARCH is not nested with any of the others due to the log-specification. Both tests rely on the loss differential $d_t = L_{1,t} - L_{2,t}$ under QML. The Diebold-Mariano tests whether the average d_t is zero via

$$DM = \frac{\bar{d}}{\sqrt{\widehat{\text{Var}}(\bar{d})}} \;\sim\; N(0,1) \;\text{ under } H_0 : \mathbb{E}[d_t] = 0 \tag{16}$$

where $\widehat{\text{Var}}(\bar{d})$ is the heteroskedasticity and autocorrelation (HAC) consistent estimator for the variance of $\bar{d}$ (long run variance). For nested models such variance degenerates and the asymptotic relation breaks, which is why the Giacomini-White test is introduced. The GW test consists in testing whether d_t is predictable from its past, meaning if there exists certain AR coefficients that are different from zero, formally

$$d_t = \beta_0 + \beta_1 d_{t-1} + u_t,\; GW = \hat{\beta}'\,\widehat{\text{Var}}(\hat{\beta})^{-1}\,\hat{\beta} \;\sim\; \chi_2^2 \text{ under } H_0 :\; \beta_0 = \beta_1 = 0. \tag{17}$$

The results of the tests are reported in **Table 5** and **Table 6** and the equivalence relations are illustrated in **Figure 6**

Table 4: Out-of-sample forecast losses.

Horizon	Model	MSE	QML
1d	EWMA	24.8818	0.7383
1d	GARCH	21.3000	0.6867
1d	GJR-GARCH	21.5955	0.6580
1d	EGARCH	21.9526	0.6517
5d	EWMA	325.8703	2.4194
5d	GARCH	266.1615	2.3666
5d	GJR-GARCH	267.1842	2.3512
5d	EGARCH	267.5110	2.3676
10d	EWMA	1146.5555	3.2014
10d	GARCH	974.3641	3.1389
10d	GJR-GARCH	980.9669	3.1402
10d	EGARCH	942.6155	3.2104

Table 5: Diebold-Mariano test results.

Horizon	Model A	Model B	DM statistic	p-value	Significant difference
1d	EWMA	EGARCH	3.406	0.0010	yes
1d	GARCH	EGARCH	2.299	0.0220	yes
1d	GJR-GARCH	EGARCH	0.625	0.5320	no
5d	EWMA	EGARCH	1.830	0.0670	no
5d	GARCH	EGARCH	-0.038	0.9690	no
5d	GJR-GARCH	EGARCH	-0.662	0.5080	no
10d	EWMA	EGARCH	-0.146	0.8840	no
10d	GARCH	EGARCH	-0.909	0.3630	no
10d	GJR-GARCH	EGARCH	-1.011	0.3120	no

Table 6: Giacomini-White test results.

Horizon	Model A	Model B	GW statistic	p-value	Significant difference
1d	EWMA	GARCH	8.505	0.0140	yes
1d	EWMA	GJR-GARCH	8.441	0.0150	yes
1d	GARCH	GJR-GARCH	4.733	0.0940	no
5d	EWMA	GARCH	180.796	< 0.0001	yes
5d	EWMA	GJR-GARCH	231.533	< 0.0001	yes
5d	GARCH	GJR-GARCH	268.892	< 0.0001	yes
10d	EWMA	GARCH	1188.073	< 0.0001	yes
10d	EWMA	GJR-GARCH	938.167	< 0.0001	yes
10d	GARCH	GJR-GARCH	213.642	< 0.0001	yes

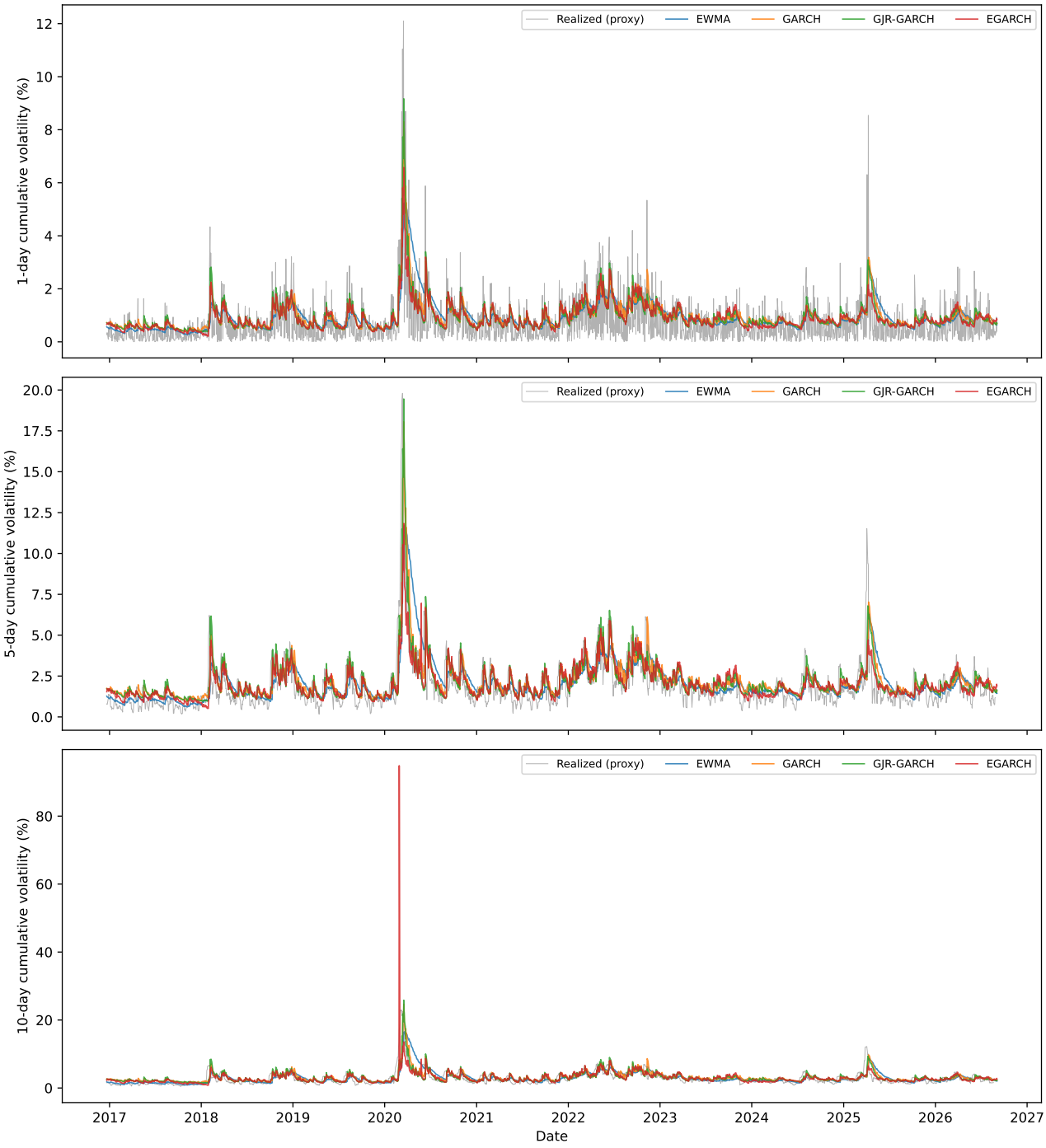


Figure 5: 1, 5 and 10-days-ahead volatility forecasts for EWMA, GARCH, GJR-GARCH and EGARCH against the realized proxy.

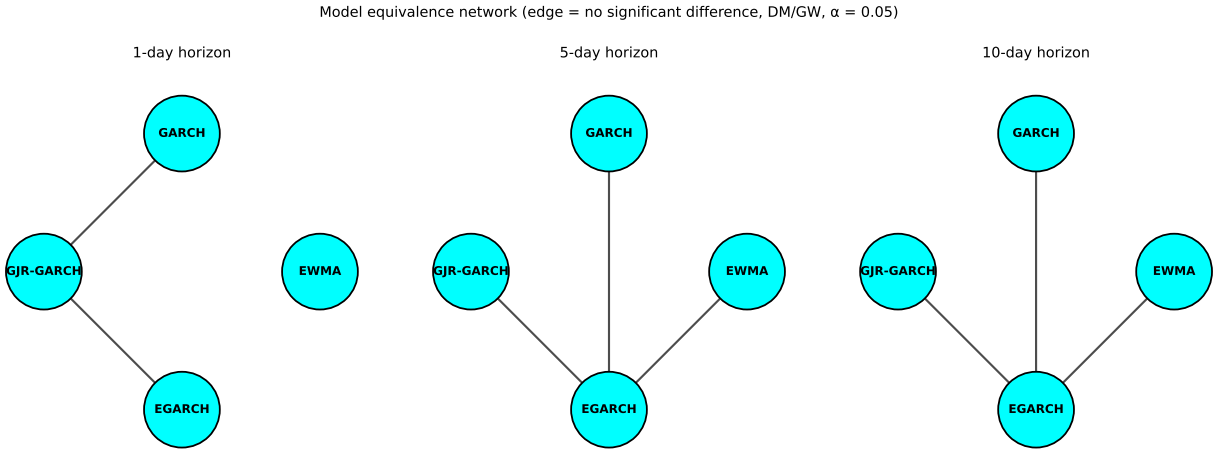


Figure 6: Illustration of the equivalence relation in the models' forecasts across all horizons.

3.3 Value-at-Risk

The Value-at-Risk (VaR) is by definition the α-quantile of the loss at a given horizon. 1, 5 and 10 days VaR are forecasted at both the 95% and 99% confidence levels for all of the three models subject of our study, both under normal and student-t innovation distribution. The VaR is computed as

$$\mathrm{VaR}_{t,h} = h\,\hat{\mu}_t + \sqrt{\widehat{\Sigma}_{t,h}}\; q_\alpha \tag{18}$$

where $q_\alpha = \Phi^{-1}(\alpha)$ for the normal case and $q_\alpha = \frac{t_\nu^{-1}(\alpha)}{\sqrt{\nu/(\nu-2)}}$ for Student-t. The forecasts are confronted against the "Historical Simulation" (HS) VaR computed as the empirical α-quantile of h-day returns sums drawn from a rolling 1000 day window, formally

$$\mathrm{VaR}^{\mathrm{HS}}_{t,h} = \mathrm{Percentile}_\alpha\big(\{r_i + r_{i+1} + \cdots + r_{i+h-1}\}_i\big). \tag{19}$$

The various forecast results are illustrate in **Figure 7** and **Figure 8** where one can appreciate the far more conservative nature of the 99% confidence forecasts with respect to the 95% ones.

The results of the VaR forecasts are tested via Kupiec unconditional coverage test [11] to check if the empirical violation rate of the VaR is consistent with α and Christoffersen independence test [4] to check if violation tend to cluster for a good VaR model should have independent violations. Both tests are Likelihood Ratios (LR). The conditional coverage test is simply the combination of the precedents.

The Kupiec unconditional coverage test is given by

$$LR_{uc} = -2\ln\left[\frac{(1-\alpha)^{n-x}\alpha^x}{(1-\hat{\pi})^{n-x}\hat{\pi}^x}\right] \;\sim\; \chi^2_1 \text{ under } H_0 : \pi = \alpha \tag{20}$$

with $\hat{\pi} = x/n$ is the observed violation rate, x is the number of violation, n the total number of observation.

The Christoffersen independence test is given by

$$LR_{ind} = -2\ln\left[\frac{(1-\pi)^{n_{00}+n_{10}}\pi^{n_{01}+n_{11}}}{(1-\pi_{01})^{n_{00}}\pi_{01}^{n_{01}}(1-\pi_{11})^{n_{10}}\pi_{11}^{n_{11}}}\right] \;\sim\; \chi^2_1 \text{ under } H_0 : \pi_{01} = \pi_{11} = \pi \tag{21}$$

where n_{00},n_{01},n_{10} and n_{11} are respectively the number of days with no violation followed by no violation, no violation followed by violation, violation followed by no violation and violation followed by violation and π_{01} and π_{11} are respectively the probability of a violation today given no violation yesterday or given a violation yesterday.

The conditional coverage test is simply given by the combination of the two

$$LR_{cc} = LR_{uc} + LR_{ind} \;\sim\; \chi^2_2 \text{ under } H_0 : \pi_{01} = \pi_{11} = \pi = \alpha \tag{22}$$

and one would ideally want a VaR model to specifically pass this last test.

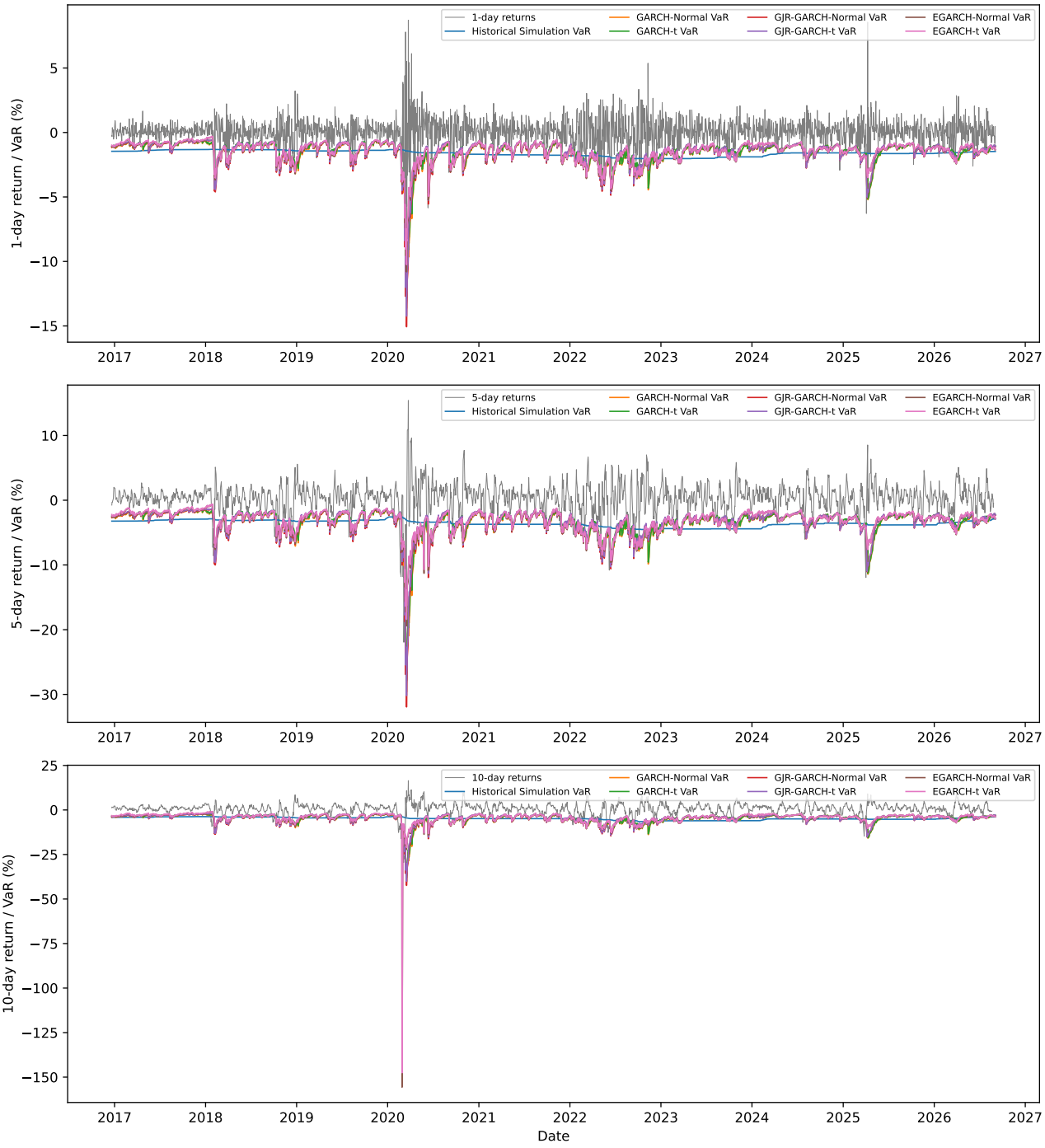


Figure 7: 1, 5, and 10-day-ahead 95% VaR forecasts all methods against realized returns.

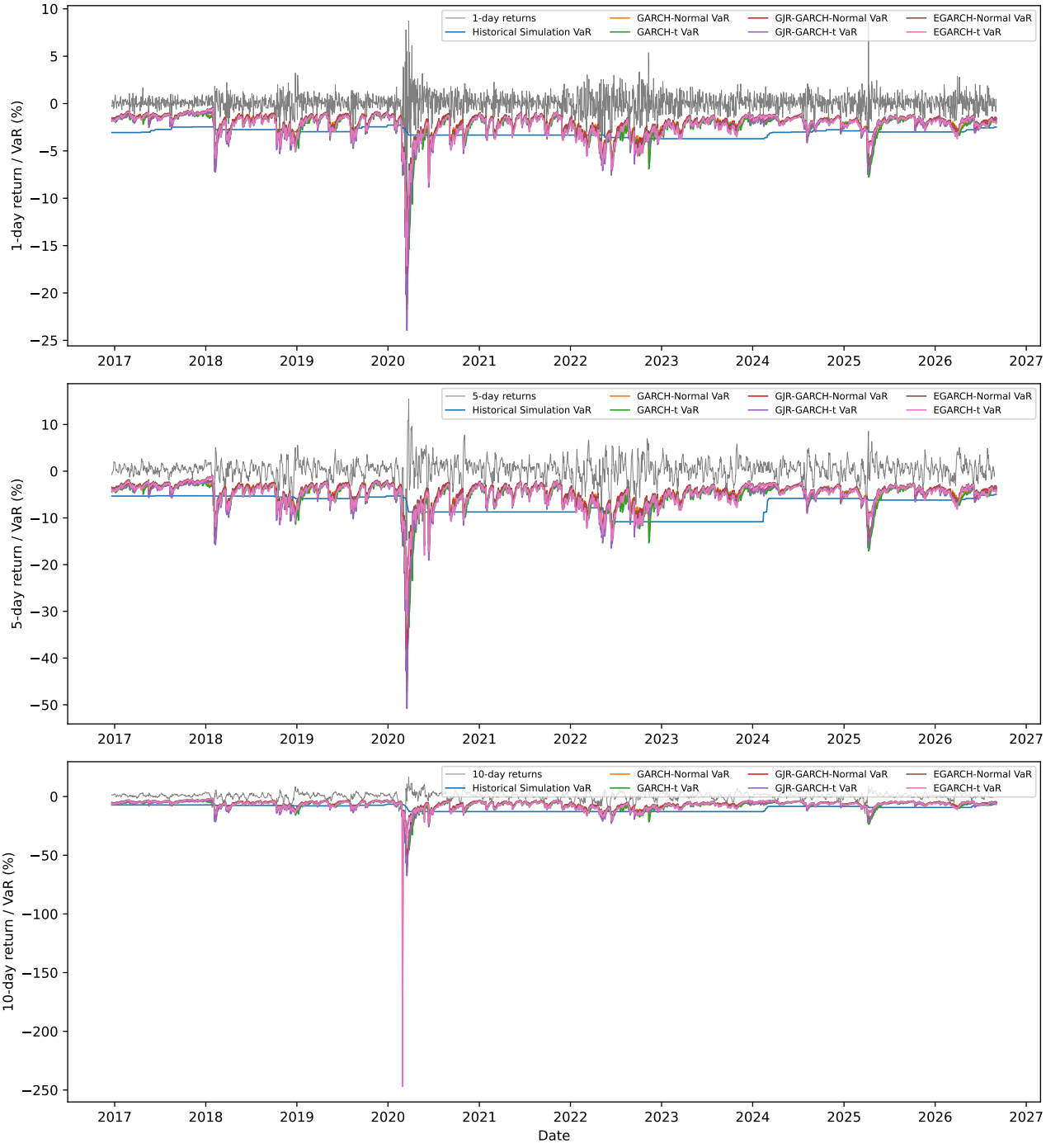


Figure 8: 1, 5, and 10-day-ahead 99% VaR forecasts all methods against realized returns.

It is relevant to note that the tests assume independent violations under but VaR forecasts at multiple days horizons are built from overlapping return windows which automatically induces the clustering of violations regardless of the model quality. To avoid this spurious influence on the results only non-overlapping forecast origins spaced $h = 5, 10$ trading days apart are used for the 5 and 10 day horizons in spite of the lower number of resulting observation and test power.

Table 7: VaR conditional independence test results for all horizons.

Model	Violations	Rate	LR_{uc}	p_{uc}	LR_{ind}	p_{ind}	LR_{cc}	p_{cc}
1 day horizon	95% Confidence Level	($n = 2437$, Target violation rate = 0.05)						
Historical Simulation	118	0.0484	0.129	0.7191	16.896	< 0.0001	17.025	0.0002
GARCH-Normal	132	0.0542	0.868	0.3516	0.108	0.7419	0.976	0.6139
GARCH-t	144	0.0591	4.015	0.0451	0.280	0.5968	4.295	0.1168
GJR-GARCH-Normal	140	0.0574	2.721	0.0990	0.000	0.9863	2.721	0.2565
GJR-GARCH-t	150	0.0616	6.397	0.0114	0.363	0.5469	6.760	0.0340
EGARCH-Normal	138	0.0566	2.165	0.1412	1.310	0.2524	3.475	0.1760
EGARCH-t	149	0.0611	5.964	0.0146	0.954	0.3286	6.918	0.0315
1 day horizon	99% Confidence Level	($n = 2437$, Target violation rate = 0.01)						
Historical Simulation	24	0.0098	0.006	0.9398	10.389	0.0013	10.395	0.0055
GARCH-Normal	50	0.0205	20.880	< 0.0001	0.761	0.3831	21.641	< 0.0001
GARCH-t	39	0.0160	7.505	0.0062	2.006	0.1567	9.511	0.0086
GJR-GARCH-Normal	49	0.0201	19.442	< 0.0001	0.845	0.3579	20.288	< 0.0001
GJR-GARCH-t	38	0.0156	6.579	0.0103	2.158	0.1418	8.737	0.0127
EGARCH-Normal	55	0.0226	28.668	< 0.0001	1.893	0.1688	30.562	< 0.0001
EGARCH-t	45	0.0185	14.115	0.0002	3.582	0.0584	17.696	0.0001
5 day horizon	95% Confidence Level	($n = 487$, Target violation rate = 0.05)						
Historical Simulation	24	0.0493	0.005	0.9419	0.524	0.4692	0.529	0.7676
GARCH-Normal	38	0.0780	6.931	0.0085	1.992	0.1582	8.923	0.0115
GARCH-t	39	0.0801	7.909	0.0049	2.236	0.1348	10.146	0.0063
GJR-GARCH-Normal	38	0.0780	6.931	0.0085	1.992	0.1582	8.923	0.0115
GJR-GARCH-t	41	0.0842	10.032	0.0015	2.766	0.0963	12.798	0.0017
EGARCH-Normal	41	0.0842	10.032	0.0015	2.766	0.0963	12.798	0.0017
EGARCH-t	41	0.0842	10.032	0.0015	2.766	0.0963	12.798	0.0017
5 day horizon	99% Confidence Level	($n = 487$, Target violation rate = 0.01)						
Historical Simulation	6	0.0123	0.247	0.6195	0.150	0.6985	0.397	0.8201
GARCH-Normal	14	0.0287	11.481	0.0007	0.831	0.3621	12.312	0.0021
GARCH-t	11	0.0226	5.744	0.0165	0.509	0.4753	6.253	0.0439
GJR-GARCH-Normal	16	0.0329	16.063	0.0001	1.090	0.2966	17.152	0.0002
GJR-GARCH-t	12	0.0246	7.489	0.0062	0.608	0.4357	8.097	0.0174
EGARCH-Normal	14	0.0287	11.481	0.0007	0.831	0.3621	12.312	0.0021
EGARCH-t	11	0.0226	5.744	0.0165	0.509	0.4753	6.253	0.0439
10 day horizon	95% Confidence Level	($n = 243$, Target violation rate = 0.05)						
Historical Simulation	11	0.0453	0.118	0.7310	0.436	0.5091	0.554	0.7581
GARCH-Normal	17	0.0700	1.823	0.1770	0.038	0.8447	1.861	0.3943
GARCH-t	17	0.0700	1.823	0.1770	0.038	0.8447	1.861	0.3943
GJR-GARCH-Normal	20	0.0823	4.506	0.0338	0.349	0.5548	4.855	0.0883
GJR-GARCH-t	21	0.0864	5.626	0.0177	0.517	0.4720	6.143	0.0463
EGARCH-Normal	19	0.0782	3.495	0.0615	0.212	0.6451	3.708	0.1566
EGARCH-t	19	0.0782	3.495	0.0615	0.212	0.6451	3.708	0.1566
10 day horizon	99% Confidence Level	($n = 243$, Target violation rate = 0.01)						
Historical Simulation	2	0.0082	0.082	0.7749	7.437	0.0064	7.519	0.0233
GARCH-Normal	6	0.0247	3.760	0.0525	2.372	0.1235	6.132	0.0466
GARCH-t	5	0.0206	2.103	0.1470	3.101	0.0783	5.204	0.0741
GJR-GARCH-Normal	7	0.0288	5.760	0.0164	1.797	0.1801	7.556	0.0229
GJR-GARCH-t	4	0.0165	0.858	0.3544	0.135	0.7139	0.992	0.6090
EGARCH-Normal	7	0.0288	5.760	0.0164	1.797	0.1801	7.556	0.0229
EGARCH-t	5	0.0206	2.103	0.1470	3.101	0.0783	5.204	0.0741

The test results are reported in **Table 7**. A p-value above 0.05 means that the model passes the specific test, meaning that it is not possible to reject in good confidence the null of unconditional coverage, independence or conditional coverage. It should be furthermore pointed out that some models' forecasts are characterized not only by the same number of violations, same LR_{uc}, but by the same LR_{ind}, likely due to isolated violations, which explains the identical rows in **Table 7**. This issue is increasingly evident for longer forecasting horizons where the sample size drops notably, weakening the power of the tests. To ensure that this is not a numerical or computational bug it has been tested that the forecast series are indeed different albeit being equally susceptible to extreme returns.

3.4 Expected Shortfall

To complete the GARCH tournament it is appropriate to add a brief comment on the expected shortfall (ES) which is by definition the conditional average realized loss on the days that saw returns breach the VaR threshold, formally

$$\mathrm{ES}_{t,h} = \mathbb{E}\big[r_{t,h} \,\big|\, r_{t,h} < \mathrm{VaR}_{t,h}\big]. \tag{23}$$

All empirical ES values are reported in **Table 8** and all models' forecasts are plotted in **Figure 9** and **Figure 10**.

Table 8: Empirical Expected Shortfall by model, confidence level and horizon.

Model	Level	$\mathrm{ES}^{\mathrm{emp}}$ (1d)	$\mathrm{ES}^{\mathrm{emp}}$ (5d)	$\mathrm{ES}^{\mathrm{emp}}$ (10d)
Historical Simulation	95%	−2.646	−6.496	−8.410
Historical Simulation	99%	−4.785	−10.923	−16.334
GARCH-Normal	95%	−2.181	−5.022	−6.630
GARCH-Normal	99%	−2.765	−7.313	−9.736
GARCH-t	95%	−2.125	−4.946	−6.630
GARCH-t	99%	−2.802	−8.304	−10.716
GJR-GARCH-Normal	95%	−2.069	−5.020	−6.189
GJR-GARCH-Normal	99%	−2.588	−7.046	−9.296
GJR-GARCH-t	95%	−2.054	−4.852	−6.013
GJR-GARCH-t	99%	−2.789	−7.884	−8.118
EGARCH-Normal	95%	−2.144	−4.840	−6.355
EGARCH-Normal	99%	−2.844	−7.172	−9.296
EGARCH-t	95%	−2.095	−4.840	−6.355
EGARCH-t	99%	−2.759	−7.890	−10.716

A Mincer-Zarnowitz regression test [13] is then executed to asses the ES forecasting performance of the various models. The regression is performed on the VaR violations

$$\Big(\mathrm{ES}^{\mathrm{mod}}_{\alpha,t,h} - r_{t,h}\Big) \cdot \mathbb{1}_{[r_{t,h} \le \mathrm{VaR}_{\alpha,t,h}]} = x_t^T \gamma + \epsilon_t \tag{24}$$

and the null $H_0 : \ \gamma = 0$ is tested with an HAC t statistic, similar to the Giacomini-White one. Failure to reject in good confidence the null hypothesis suggest that the model provides an appreciable ES forecast. The results of this test are reported in **Table 9** where it can be seen that the null can be refuted at both confidence level at the 1 day horizon for all the models with normal innovations. At longer horizons, although it must be noted that the test looses power due to the lower amount of available data, which is especially evident at the 10 days horizon 99% confidence forecasts, it can be still be seen that the student-t specified models generally provide for higher p-values and hence more significant ES forecasts. These results once again confirm that the choice of the innovation distribution is far more influential on the forecasting capabilities than the choice of the GARCH-type specification itself and that the student-t distribution better captures tail risk.

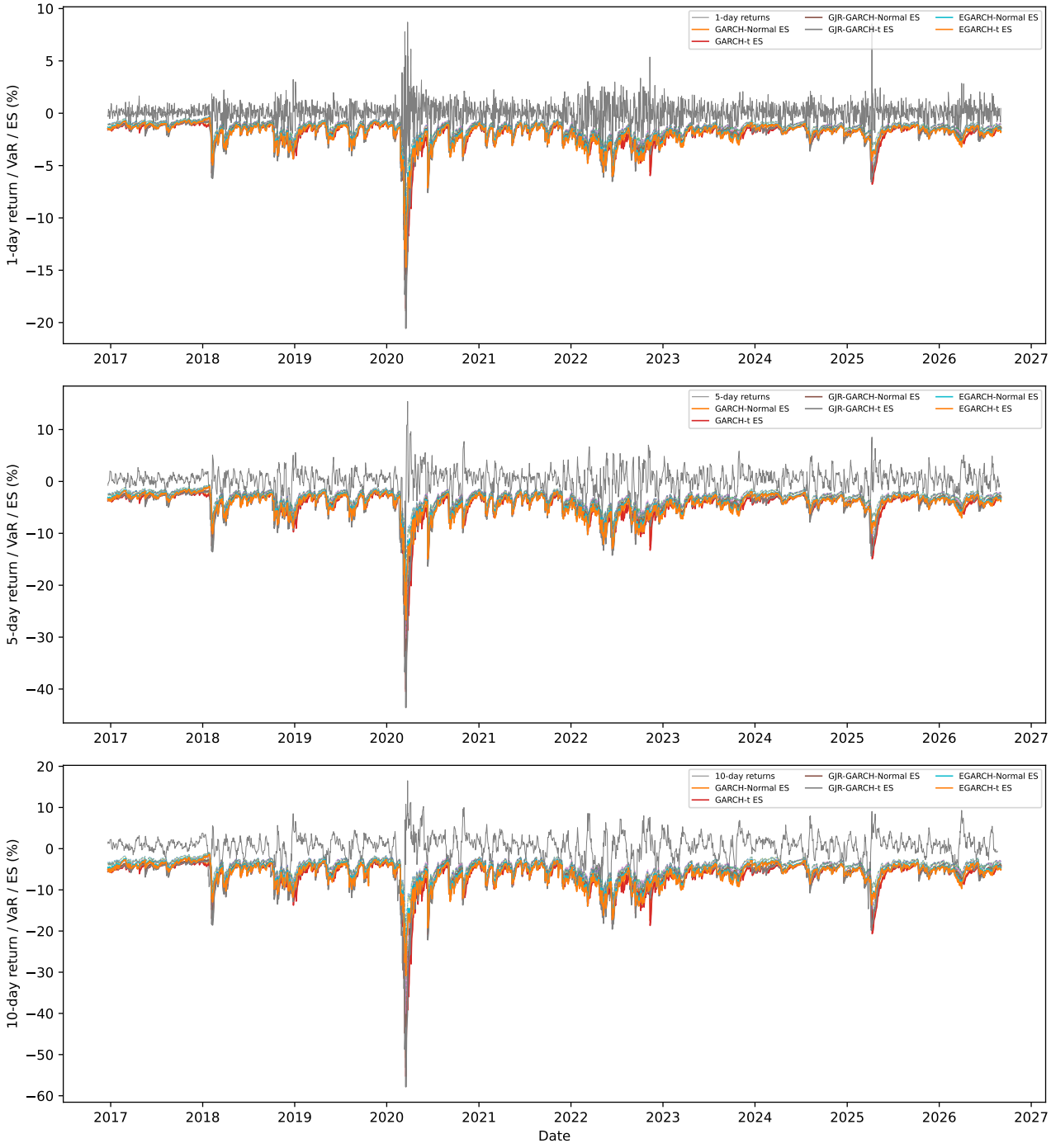


Figure 9: Expected shortfall forecast at 95% VaR confidence.

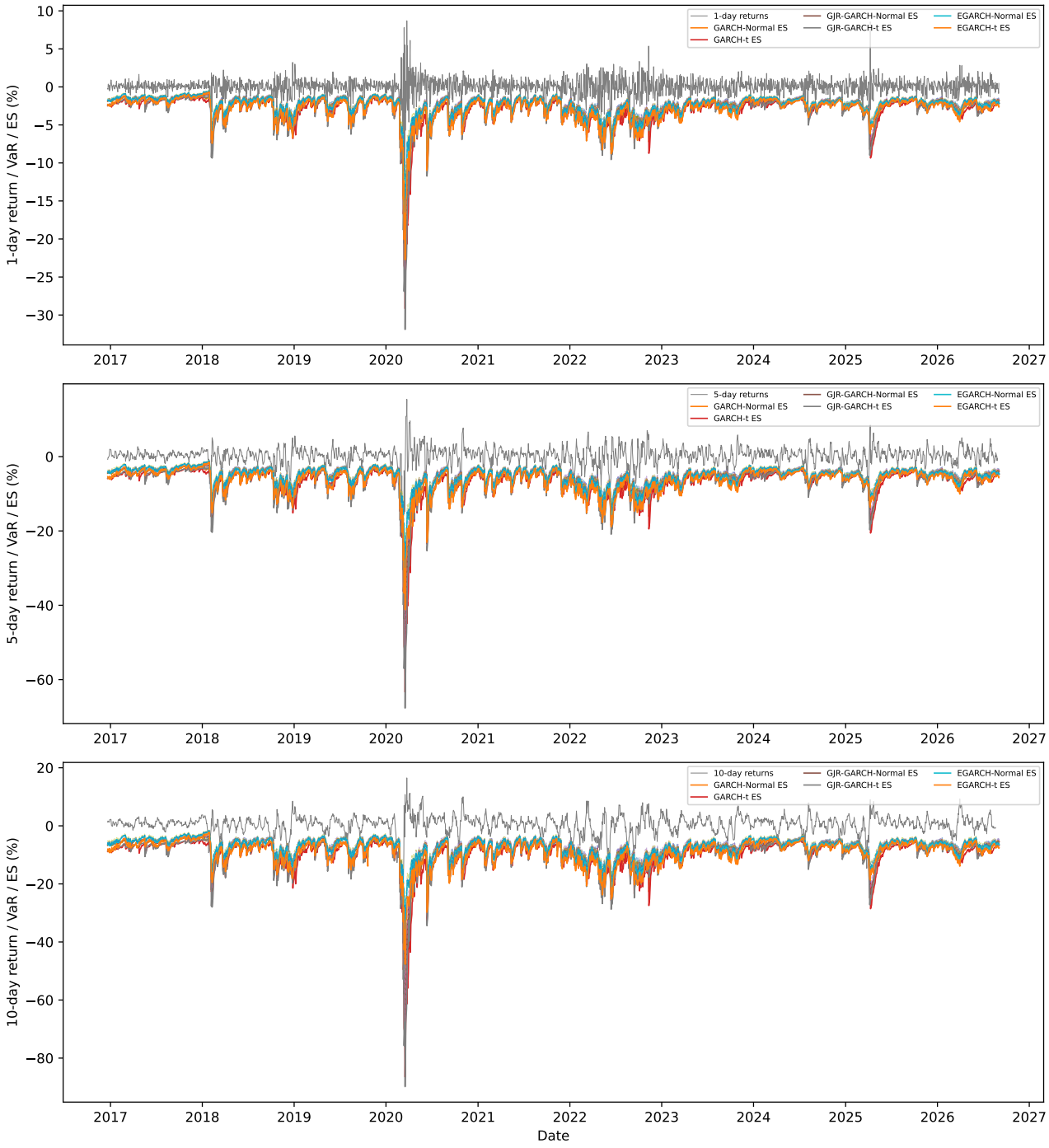


Figure 10: Expected shortfall forecast at 99% VaR confidence.

Table 9: Mincer-Zarnowitz regression test results for Expected Shortfall on all horizons.

Model	n	Exceedances	$\hat{\gamma}$	MZ statistic	p-value
	1 day horizon	95% Confidence Level			
GARCH-Normal	2437	132	0.0148	3.7144	0.0002
GARCH-t	2437	144	0.0043	1.0770	0.2815
GJR-GARCH-Normal	2437	140	0.0129	3.3109	0.0009
GJR-GARCH-t	2437	150	0.0026	0.6541	0.5130
EGARCH-Normal	2437	138	0.0171	4.0250	0.0001
EGARCH-t	2437	149	0.0073	1.8012	0.0717
	1 day horizon	99% Confidence Level			
GARCH-Normal	2437	50	0.0090	3.4785	0.0005
GARCH-t	2437	39	0.0017	0.9272	0.3538
GJR-GARCH-Normal	2437	49	0.0088	3.5211	0.0004
GJR-GARCH-t	2437	38	0.0020	1.0981	0.2722
EGARCH-Normal	2437	55	0.0096	3.5062	0.0005
EGARCH-t	2437	45	0.0018	0.8108	0.4175
	5 day horizon	95% Confidence Level			
GARCH-Normal	487	38	0.0733	1.8756	0.0607
GARCH-t	487	39	0.0461	1.2347	0.2170
GJR-GARCH-Normal	487	38	0.0850	2.1439	0.0320
GJR-GARCH-t	487	41	0.0550	1.4395	0.1500
EGARCH-Normal	487	41	0.0744	1.9157	0.0554
EGARCH-t	487	41	0.0500	1.3460	0.1783
	5 day horizon	99% Confidence Level			
GARCH-Normal	487	14	0.0546	1.8796	0.0602
GARCH-t	487	11	0.0213	1.0698	0.2847
GJR-GARCH-Normal	487	16	0.0548	1.8165	0.0693
GJR-GARCH-t	487	12	0.0233	1.1214	0.2621
EGARCH-Normal	487	14	0.0518	1.7798	0.0751
EGARCH-t	487	11	0.0238	1.1898	0.2341
	10 day horizon	95% Confidence Level			
GARCH-Normal	243	17	0.0650	1.1704	0.2419
GARCH-t	243	17	0.0324	0.6450	0.5189
GJR-GARCH-Normal	243	20	0.0729	1.4088	0.1589
GJR-GARCH-t	243	21	0.0345	0.7515	0.4523
EGARCH-Normal	243	19	0.0897	1.3603	0.1737
EGARCH-t	243	19	0.0584	0.9518	0.3412
	10 day horizon	99% Confidence Level			
GARCH-Normal	243	6	0.0353	1.1542	0.2484
GARCH-t	243	5	-0.0177	-1.4531	0.1462
GJR-GARCH-Normal	243	7	0.0322	1.2553	0.2094
GJR-GARCH-t	243	4	0.0135	0.6728	0.5011
EGARCH-Normal	243	7	0.0533	1.2191	0.2228
EGARCH-t	243	5	0.0135	0.7457	0.4558

4 Conclusions

In conclusion, as one could have expected, no single GARCH-type model, out of the observed three, emerges as the overall "winner" in that it consistently outclass all others in in-sample and out-of-sample fit and VaR modeling. Depending on the specific task, be it modeling historical data or forecasting and risk management, the required forecast confidence, time horizon of interest and computational resources available, one or the other model may prove more appropriate. It is however clear that the choice of the student-t innovations clearly outclass the normal assumption since it better captures tail risk and provides for more reliable risk measures overall.

The resulting work could be taken as a base to expand on, for instance by adding other GARCH-type models to the tournament or by studying their properties on multi-asset portfolios. Furthermore, the estimation window, the re-estimation intervals and the forecast have all been rather arbitrarily set and could be subject of further inquiries. Other nuances that can be investigated relate to the innovations' distributional assumptions, for example Cornish-Fisher approximation for adjusted quantiles and asymmetric Student-t distribution could be explored.

The approach towards this project has been mainly didactic and aimed at a better comprehension of the course material albeit some little expansions on it are present. The entire project focused only on the second half of the course, personally deemed less intuitive than the first, and as previously mentioned in **Subsection 2.4** the constant mean assumption obviously proves problematic, especially further down the line for what concerns the evaluation of the loss functions. Indeed, to obtain a complete picture one should combine ARMA and GARCH-type models so as to study both mean and volatility behavior.

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